



Youssef Hadjkacem

✉ youssef.hadjkacem25@gmail.com [in](#) LinkedIn [G](#)itHub

Final-year computer engineering student at ENSI, who deployed a **FastAPI** API to production on **AWS** (EC2, S3, IAM) during an internship at Yonnov'IA. Seeking a 4-to-6-month internship to deepen expertise in **microservices** architecture and **CI/CD** pipelines.

Education

National School of Computer Science (ENSI) Engineering Degree – Computer Science (Final Year)	Sept. 2024 – 2027 Tunis, Tunisia
Faculty of Sciences of Sfax Preparatory Classes – Mathematics (Rank: 110/1800)	June 2022 – June 2024 Sfax, Tunisia

Professional Experience

Backend/AI Intern June 2026 – August 2026	Yonnov'IA Remote, France
– Sales teams lacked a reliable tool to generate quotes ; AI copilot leveraging sales history and CRM data, integrated into an Odoo ERP via a FastAPI REST API, deployed on AWS (EC2, S3, IAM) with an MCP interface for interoperability with external AI agents. – Technologies: Python, FastAPI, Ollama, AWS, MCP	
AI Intern, Biomechanics & Exoskeleton June 2026 – August 2026	University of Lorraine Remote, France
– A reinforcement-learning exoskeleton controller needs a faithful biomechanical model ; 3D digital twin and musculoskeletal analysis (OpenSim), preparing MuJoCo/MyoSuite integration via isolated Docker environments, with an automated validation pipeline cutting the anomaly rate from 30% to 0%. – Technologies: Python, Docker, OpenSim, MuJoCo	
Software Engineering Intern June 2025 – August 2025	Fogits Sfax, Tunisia
– Manually entering LinkedIn leads was slow and error-prone ; Python scraper (Selenium + BeautifulSoup) processing 800+ profiles, loaded via Docker into an Odoo ERP (-85% manual entry). – Technologies: Python, Selenium, Docker, Odoo	

Projects

AI Emergency Savior 🤖

- Recommending an emergency care provider requires balancing real-time constraints ; **FastAPI REST API** orchestrating geolocation, OCR and a **multi-objective optimization engine (NSGA-II)**, containerized (Docker) on a FastAPI + React + PostgreSQL stack for reproducible deployment.
- **Technologies:** FastAPI, React, TypeScript, Docker, PostgreSQL

GenAI Pipeline & Industrial RAG System 🤖

- Reliably answering business questions requires real sources ; **Django REST API** orchestrating an **autonomous agent** (LangGraph, ReAct) on a **local LLM** (Ollama) in an isolated container, with **FAISS** vector indexing.
- **Technologies:** Django REST, FAISS, Ollama, Docker

Microservices API with CI/CD

- A monolithic service was becoming hard to evolve ; split into **FastAPI microservices** communicating via **Redis**, orchestrated with **Docker Compose** and a **CI/CD** pipeline (GitHub Actions).
- **Technologies:** FastAPI, Docker Compose, GitHub Actions, Redis

Technical Skills

Languages: Python, C, C++, Java	Backend & API: FastAPI, Django REST, Flask, Odoo/CRM
Databases: PostgreSQL, MongoDB, Redis	Containerization: Docker, Docker Compose
CI/CD: GitHub Actions, Kubernetes	Cloud: AWS (EC2, S3, IAM)

Languages

Arabic: Native	English: C1	French: B2
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Leadership & Volunteering

ENSI Junior Enterprise (EJE) – International Relations Officer	Tunis, Tunisia
AIESEC – Python Teacher	Târgu Jiu, Romania
AIESEC – English Teacher	Gaziantep, Turkey